Assignment 3: RetailTech Analytics Challenge

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Introduction

This assignment works with the RetailTech review dataset to solve several specific business requirements. To obtain the relevant review data, we utilize the Amazon Review dataset collected in 2023 by McAuley Lab, which contains product reviews from customers, including ratings, review text, helpful vote and associated metadata. Five categories related to the companies' operation are used for analysis, including *Electronics, Books, Home_and_Kitchen, All_Beauty, Clothing_Shoes and_Jewelry.*

Challenge 1, 2, 4 ,5 are chosen to be completed. Objectives for these challenges are to use the selected review data to address product categorization, rating prediction and custom logistic model implementation

Challenge 1: Product Categorization

Methodology and Results

The goal of the first challenge is to build a model that could predict the product category based on the review text content. For each of the five categories, 5000 review samples are drown for the task. In total, our dataset contains 25000 reviews. Based on Exercise2, pre-trained transformer *all-MiniLM-L6-v2* is applied to conduct text embeddings.

Since single reviews can be noisy or sparse, so aggregating at the product level helps capture a more complete picture of a product. Thus, for Challenge 1, where we are trying to classify the product (not just the review) into categories, review embeddings are integrated to product level under mean aggregation method. The dataset is then split into training and test set before modeling.

Firstly, we trained *KNN, Logistic Regression* and *Random Forest* classifier with stratified 5-fold cross-validation. As shown in Table 1, the performance for all three is under the requirement 85% accuracy, the F1 score balances precision and recall, and the weighted version accounts for class imbalance by averaging F1s per class, weighted by the number of instances. F1 all align closely with accuracy, suggests good performance.

Table 1 Cross Validation Summary

	Mean Accuracy	F1 (weighted)
KNN	0.825 ± 0.007	0.825 ± 0.007
Logistic Regression	0.843 ± 0.007	0.843 ± 0.007
Random Forest	0.835 ± 0.005	0.835 ± 0.005

In order to achieve our accuracy target, *GridSearchCV* is conducted to finetune the hyperparameter for logistic regression as it performed best among the three models. The best logistic model reached 85% overall accuracy. Additionally, an ensemble model (logistic + random forest) is trained with cross validation evaluation, and performed the best. As summarized in Figure 1, the cross validation

average accuracy is above 85.1% and with a very small volatility. For each category, it generally classifies well.

	precision	recall	f1-score	support
All_Beauty	0.85	0.80	0.82	744
Books	0.94	0.95	0.94	979
Clothing_Shoes_and_Jewelry	0.86	0.86	0.86	961
Electronics	0.83	0.86	0.85	902
Home_and_Kitchen	0.79	0.80	0.79	959
accuracy			0.85	4545
macro avg	0.85	0.85	0.85	4545
weighted avg	0.85	0.85	0.85	4545

CV Accuracy: 0.851 ± 0.004

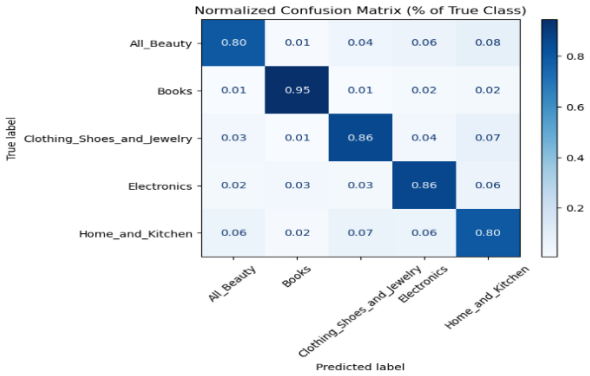

Figure 1 Classification Report and Confusion Matrix for Ensemble Model

From confusion matrix we can tell the classifier works better for *Books, Clothing and Electronics*. For *Beauty, Home_and_Kitchen* products it still gets confused sometimes. In test set which contains 4545 instances, 660 samples in total are misclassified. Most common error made by the model is to misclassify *Clothing_Shoes_and_Jewelry* as *Home_and_Kitchen* (70 cases). Table 2 summarizes the top 5 mismatch pairs. *Home_and_Kitchen* appears in most of the misclassification cases, which aligns with the result from the classification report and confusion matrix.

Table 2 Top Misclassification Pairs

True Label	Predicted	Number of Cases
Clothing_Shoes_and_Jewelry	Home_and_Kitchen	70
Home_and_Kitchen	Clothing_Shoes_and_Jewelry	65
All_Beauty	Home_and_Kitchen	63
Home_and_Kitchen	Electronics	59
Home_and_Kitchen	All_Beauty	54

To try to understand what caused the confusion, PCA plot is utilized to project review embeddings into lower dimension. As shown in Figure 2, we can see *Books* cluster and *Clothing_Shoes_and_Jewelry* cluster are separated well. While *Home_and_Kitchen* overlays other categories, which explains the high misclassification rates.

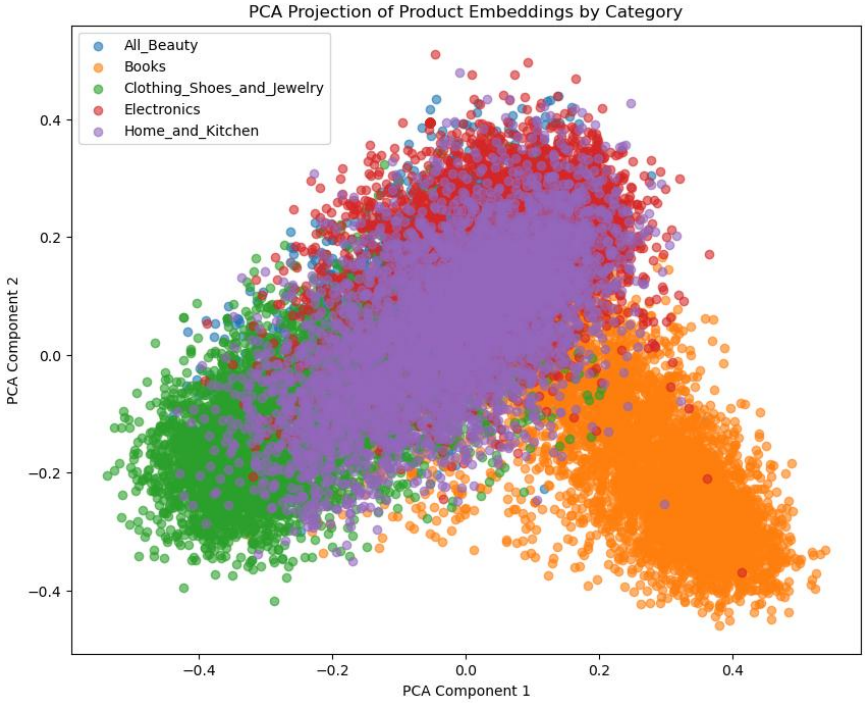

Figure 2 PCA Projection of Product Embeddings by Category

Analysis

The current transformer model used (all-MiniLM-L6-v2) was trained on general text. For product department, fine-tuning it on their domain-specific data may produce embeddings that are more discriminative for categories related to our business. Also combining transformer-based embeddings with traditional methods like TF-IDF vectors or incorporating metadata (such as review length, verified purchase status, etc.) might capture complementary aspects that improve classification.

To get a better error control, product department may consider develop mechanisms where misclassified products can be manually reviewed and then used to refine the categorization rules or further train the model. This feedback loop can improve performance over time and help uncover edge cases.

Challenge 2: Rating Prediction

Methodology and Results

Challenge 2 aims to predict numerical rating using review text. Both regression and classification models are trained for comparison. The same dataset and corresponding embeddings are utilized for the task. The initial attempt using raw embedding yielded undesirable result. Only 47.3% predicted reviews are within 0.5 stars of the actual rating with a mean absolute error of 0.679. So further feature engineering is needed for improvement.

We applied *SentimentIntensityAnalyzer* to acquire sentiment scores, together with review length and log normalized helpfulness vote, adding new features before the modelling. Furthermore, since the embeddings have 384 dimensions, PCA transform is applied to reduce the embedding dimension. The dataset is then split into training and test set again. For regression task, Linear Regression and SVR are implemented, along with cross validation and GirdSearchCV. The best regression model could

predict 81.1% reviews within 1 stars of their actual rating, with a mean absolute error of 0.595. A logistic classification model with GirdSearchCV produces 70% accuracy.

Analysis

The results from this part reveal that while raw review embeddings provide a useful representation of review content, they are insufficient on their own to accurately predict user ratings. The addition of sentiment scores, review length, and helpfulness votes significantly enhanced model performance, suggesting that structural and emotional cues in review text play a critical role in user satisfaction. The analysis enables RetailTech to flag reviews where predicted sentiment significantly diverges from the reported rating, which may indicate: misleading ratings, poorly written reviews, or customer dissatisfaction despite a high score.

Challenge 4: Logistic Regression from Scratch

Methodology and Results

The challenge 4 requests implementation of logistic regression to deepen algorithmic understanding and allow future customization. This section details the methodology and evaluation of a custom-built logistic regression classifier designed for both binary and multi-class classification, with support for L1/L2 regularization and softmax or one-vs-rest strategies.

We developed a custom logistic regression classifier, *LogisticRegressionCustom*, using NumPy for numerical computation and Matplotlib for visualization. The implementation is modular and extensible, supporting both binary and multi-class classification through either a one-vs-rest (OvR) scheme or a softmax-based approach. Optimization is performed using batch gradient descent with a configurable learning rate and number of iterations, allowing for fine-tuned training control. The class includes optional L1 (Lasso) and L2 (Ridge) regularization mechanisms to prevent overfitting, with adjustable regularization strength. Loss computation incorporates both the prediction error and regularization penalties, using binary cross-entropy for binary tasks and categorical cross-entropy for softmax-based multiclass tasks. To monitor convergence and training stability, the model includes a plot_loss() function that visualizes the loss over iterations.

To compare this customized classifier with Scikit-learn’s Logistic Regression, the model is tested on the task for challenge 1, which aims to classify product categories. All preprocessing steps remaining identical, the customized logistic regression performs poorly, only obtained overall 64% accuracy, with a weighted F1-score of 0.58. While it performed well in certain classes (e.g., precision of 0.93 for class 3), it struggled with others, notably class 0 where both precision and recall were 0, suggesting difficulty learning class-specific features. In comparison, scikit-learn’s LogisticRegression achieved a significantly higher accuracy of 85%, benefiting from advanced optimization algorithms and internal regularization tuning.

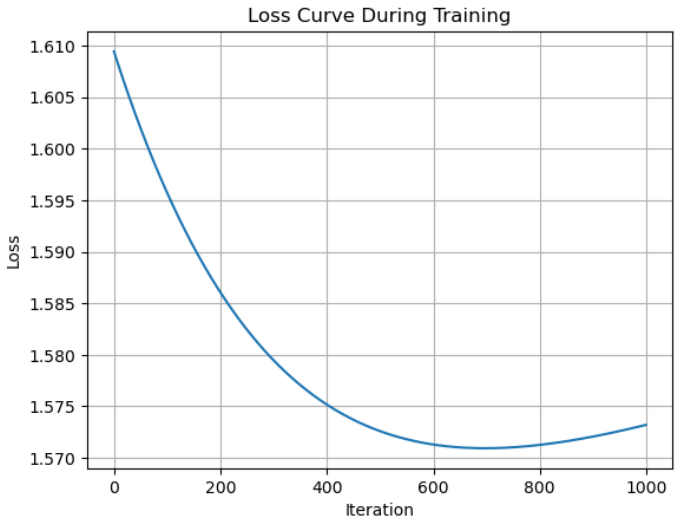

Figure 3 Loss Curve for LogisticRegressionCustom

The loss curve of the custom model, shown in Figure 3, indicated convergence during early training but showed signs of underfitting or numerical instability toward later iterations. This comparison highlights the custom model's value for educational purposes and interpretability, though it currently lags behind optimized libraries in terms of raw predictive performance.

Analysis

Extending binary logistic regression to multi-class classification introduces several challenges, both conceptual and computational. The binary model, which predicts probabilities between two classes using the sigmoid function, must be adapted to handle multiple classes simultaneously. This requires either a One-vs-Rest (OvR) strategy—training a separate binary classifier for each class—or the use of a softmax-based approach that models the joint probability distribution across all classes. OvR can be inefficient and may lead to inconsistencies when multiple classifiers assign high probabilities to different classes, while softmax requires calculating exponentials and normalizing over all class scores, increasing computational cost and sensitivity to numerical instability. Additionally, class imbalance becomes more problematic in the multiclass setting, as certain classes may dominate the loss and skew the learning process if not properly handled. These challenges make it crucial to carefully select the multiclass strategy based on dataset characteristics and implementation goals.

Challenge 5: Executive Team Challenge

The ROI is calculated assuming an initial development phase involving a small team consisting of two data scientists and one machine learning engineer working over a three-month period to build and deploy the classification systems. These models are designed to automate key tasks such as product categorization and review rating prediction, significantly reducing the need for manual intervention. Additionally, by improving the accuracy of product recommendations, the models are expected to boost conversion rates by approximately 2–3% in select product categories. Annual maintenance costs account for periodic model retraining and upkeep of supporting data pipelines. Notably, this ROI estimate is conservative—it does not factor in long-term scalability benefits, reputational gains from more trustworthy reviews, or customer loyalty effects that may accrue over time. Table 3 demonstrates the ROI analysis, showing a breakeven point of approximately 8 months and a 172%

medium-term benefit.

Table 3 ROI Calculation Worksheet – RetailTech Classification System

Category	Estimated Value (Annual)	Notes / Assumptions
Costs		
Model Development (one-time)	$40,000	Includes salaries for data scientists, engineering, and project management
System Integration	$15,000	Backend integration with product database and review pipeline
Annual Maintenance	$10,000	Bug fixes, retraining, hardware, monitoring
Cost of False Positives/Negatives	$8,000	Estimated based on business loss from misclassification (e.g., hiding good reviews)
Total Annual Cost	**$73,000**	
Benefits		
Reduced Manual Review Costs	$30,000	Assumes 2 full-time content reviewers saved (approx. $15K each annually)
Increased Sales via Better Recommendations	$50,000	Due to improved product targeting and fewer fake reviews misleading buyers
Improved Customer Satisfaction (Retention)	$20,000	Based on 2% increase in retention from clearer product review trends
Efficiency Gains in Product Categorization	$15,000	Faster onboarding of new products; fewer misclassified items
Total Annual Benefit	**$115,000**	
Net Annual Gain	**$42,000**	$115,000 - $73,000
Break-Even Point	**~8 months**	$73,000 cost / ($115,000 / 12 months)
3-Year ROI	**172%**	Total Gain: ($42,000 * 3) / Initial Investment ($73,000)